

CO1 - Assessment Tool 2

01) Full - duplex voice channels.

A) Given

Total spectrum = 45 MHz

simple channel bandwidth = 25 kHz

No. of simplex channels = $\frac{\text{Total bandwidth}}{\text{channel bandwidth}}$

No. of full duplex channels = $\frac{\text{simplex channels}}{2}$

convert MHz to kHz:

$$45 \text{ MHz} = 45000 \text{ kHz}$$

$$45000 \div 25 = 18000$$

full - duplex channels

$$18000 \div 2 = 9000$$

2) channels Per cell

Given

Total channels = 560

cluster size = 7

channels Per cell = $\frac{\text{Total channels}}{\text{cluster size}}$

$$560 \div 7 = 80$$

3) GSM voice channels

Given

No. of carriers = 100

Time slots Per carrier = 8

Total voice channels = carriers $\times$ time slots

$$100 \times 8 = 800$$

4) Power control ratio

Given :

Base station Power = 80 W

Handset Power = 4 W

$$\text{Power ratio (dB)} = 10 \log_{10} (P_1/P_2)$$

$$= 10 \log_{10} (80/4)$$

$$= 10 \log_{10} (20)$$

$$= 10 \times 1.301$$

$$= 13.01 \text{ dB} = \approx 13 \text{ dB}$$

5) carrier - to - interference Ratio

Given

Signal = -60 dBm

interference = -92 dBm

$$C/I = \text{Signal} - \text{Interference}$$

$$= (-60) - (-92) = 32 \text{ dB}$$

6) channels per cell

Given

Spectrum = 42 MHz

channel bandwidth = 300 kHz

cluster size = 1

$$42 \text{ MHz} = 42000 \text{ kHz}$$

$$\text{Total channels} = 42000 \div 300 = 140$$

$$= 140 \div 7 = 20$$

7) Received Power after doubling distance

A) Given

Initial Power = 64 W

Distance doubled

$$\text{Path loss exponent} = 4$$

$$= 64 \div 16 = 4 \text{ W}$$

8) Increase transmit Power

Given

cell radius doubled

Inverse - square law ($n=2$)

Power increase = $(2^2) = 4$

Transmit Power must increase by factor 4

a) Total channels in cluster

Given

cluster size = 4

channels per cell = 15

total channels = cluster size $\times$ channels per cell

$$= 4 \times 15 = 60$$

60 channels,,

10) Doppler shift

Given

speed = 100 km/h

frequency = 900 MHz

convert speed

$$100 \text{ km/h} = 100 \times 1000 / 3600 = 27.78 \text{ m/s}$$

$$f_d = v f / c$$

$$v = 27.78 \text{ m/s}$$

$$f = 900 \times 10^6 \text{ Hz}$$

$$c = 3 \times 10^8 \text{ m/s} = 83.34 \text{ Hz,,}$$

11) Duration one TDMA time slot

A) Given

frame duration = 5 ms

No. of slots = 8

$$\text{slot duration} = \frac{\text{frame duration}}{\text{Number of slots}}$$

$$5/8 = 0.625 \text{ ms or } 625 \mu\text{s},$$

2) Number of GSM carrier frequencies
Given

$$\text{Total bandwidth} = 24 \text{ MHz}$$

$$\text{Carrier spacing} = 200 \text{ kHz}$$

$$24 \text{ MHz} = 24000 \text{ kHz}$$

$$\text{carriers} = \frac{\text{Total bandwidth}}{\text{carrier spacing}}$$

$$= \frac{24,000}{200} = 120$$

120 carrier frequencies,

13) Received Power after Path loss

Given

$$\text{Transmit Power} = 8 \text{ W}$$

$$\text{Path loss} = 33 \text{ dB}$$

convert 8W to dBm

$$P(\text{dBm}) = 10 \log_{10}(8000) = 39.03 \text{ dBm}$$

Received Power;

$$P_r = P_t - \text{Path loss}$$

$$= 39.03 - 33 = 6.03 \text{ dBm},$$

14) Free space Path loss

Given

$$\text{frequency} = 1900 \text{ MHz}$$

$$\text{Distance} = 2.5 \text{ km}$$

$$\text{FSPL} = 32.44 + 20 \log_{10}(f \text{ MHz}) + 20 \log_{10}(d \text{ km})$$

$$\text{FSPL} = 32.44 + 20 \log_{10}(1900) + 20 \log_{10}(2.5)$$

$$= 32.44 + 65.58 + 7.96$$

$$= 105.98 \text{ dB} \approx 106 \text{ dB},$$

15) cluster call - handling - handling capacity

Given

No. of cells = 6

calls per cell = 250

Total capacity = $6 \times 250 = 1500$

1500 simultaneous calls

16) Maximum active calls.

Given

New call requests = 4000 calls/sec

Average call duration = 3 minutes

3 minutes = 180 seconds

Active calls = $\lambda \times T$

$$= 4000 \times 180 = 720,000$$

17) Total bandwidth consumed

Given

No. of channels = 150

channel bandwidth = 25 KHz

$$B = N \times BW$$

$$= 150 \times 25 = 3750 \text{ KHz}$$

$$= 3.75 \text{ MHz}$$

18) Bandwidth wasted by guard bands

A) Given

Guard band = 40 KHz

No. of guard bands = 30

wasted band width = 30×40

$$= 1200 \text{ KHz}$$

$$= 1.2 \text{ MHz}$$

19) Number of handoffs in one hour.

A) Given

$$\text{Speed} = 90 \text{ km/h}$$

$$\text{cell radius} = 1.5 \text{ km}$$

$$\text{cell diameter} = 2R = 2(1.5) = 3 \text{ km}$$

$$\text{Distance travelled in 1 hour} = 90 \text{ km}$$

$$\text{Hand offs} = 90/3 = 30$$

Approximately 30 handoffs.

20) Total user capacity

A) Given

$$\text{Regional centers} = 8$$

$$\text{Base stations per region} = 40$$

$$\text{users per base station} = 250$$

$$\text{Total users} = 8 \times 40 \times 250 = 80,000 \text{ users.}$$

21) Number of cells required

A) Given

$$\text{Total area} = 400 \text{ km}^2$$

$$\text{cell radius} = 400 \text{ m} = 0.4 \text{ km}$$

$$\text{Approximate cell area} = R^2$$

$$\text{cell Area} = R^2$$

$$\text{cell area} = (0.4)^2 = 0.16 \text{ km}^2$$

$$\text{No. of cells} = \frac{400}{0.16} = 2500 \text{ cells.}$$

22) Number of FDMA channels

A) Given

$$\text{Total spectrum} = 6 \text{ MHz}$$

$$\text{channel bandwidth} = 30 \text{ kHz}$$

$$\begin{aligned}\text{channels} &= \frac{\text{Total bandwidth}}{\text{channel bandwidth}} \\ &= \frac{600}{30} = 200 \text{ channels,}\end{aligned}$$

23) Offered traffic (Erlangs)

$$\text{subscribers} = 1200$$

$$\text{calls per subscriber} = 4 \text{ calls/hour}$$

$$\text{Average call duration} = 2.5 \text{ minutes}$$

$$2.5/60 = 0.04167 \text{ hour}$$

$$A = N \times \lambda \times h$$

$$= 1200 \times 4 \times 0.04167$$

$$= 200 \text{ erlangs,}$$

24) Maximum simultaneous users

$$\text{Total bandwidth} = 40 \text{ MHz}$$

$$\text{Bandwidth per user} = 400 \text{ KHz}$$

$$40 \text{ MHz} = 40,000 \text{ KHz}$$

$$\text{users} = \frac{40,000}{400} = 100 \text{ users,}$$

25) Total network user capacity

$$\text{users per base station} = 600$$

$$\text{No. of base stations} = 1200$$

$$\text{Total users} = 600 \times 1200 = 720,000 \text{ users.}$$

26) Paging rounds

Given

$$\text{Total cells} = 200$$

$$\text{cells searched per round} = 4$$

$$= 200/4 = 50$$

28) Roaming subscribers

Given

$$\text{Total subscribers} = 600,000$$

$$\text{Roaming Percentage} = 15\%$$

$$\text{Roaming users} = 600,000 \times 0.15 = 90,000$$

29) switching center utilization

Given

$$\text{Capacity} = 12,000 \text{ calls/sec}$$

$$\text{current load} = 9,000 \text{ calls/sec}$$

$$\text{utilization} = \frac{9,000}{12,000} \times 100 = 75\%$$

30) channels per cell

Given

$$\text{Total spectrum} = 48 \text{ MHz}$$

$$\text{Channel bandwidth} = 200 \text{ kHz}$$

$$\text{Reuse factor} = 4$$

$$48 \text{ MHz} = 48,000 \text{ kHz}$$

$$= \frac{48,000}{200} = 240 = 60$$

31) Number of GSM channels

Given

$$\text{Total spectrum} = 30 \text{ MHz}$$

$$\text{channel spacing} = 200 \text{ kHz}$$

$$30 \text{ MHz} = 30,000 \text{ kHz}$$

$$\text{channels} = \frac{\text{Total bandwidth}}{\text{channel spacing}}$$

$$\frac{30,000}{200} = 150 \text{ channels}$$

32) successful handoffs:

Given

Hand off attempts = 2500.

Success rate = 99%

$$= 99/100 \times 2500$$

$$= 2475 \text{ successful handoff}$$

33) Received Power at the base station

A) Given

Transmit Power = 12 W

Path loss = 45 dB

$$P_t (\text{dBm}) = 10 \log_{10} (12/0.001) = 40.79 \text{ dBm}$$

$$P_r = P_t - \text{Path loss} = 40.79 - 45$$

$$= -4.21 \text{ dBm} \approx -4.2 \text{ dBm}$$

34) cluster user capacity

Given

cluster size = 7 cells

user per cell = 900

$$\text{Total users} = 7 \times 900$$

$$= 6300 \text{ users}$$

35) TDMA slot duration:

Given

frame duration = 4.8 ms

slots = 8

$$\text{slot duration} = \frac{\text{frame duration}}{\text{slots}}$$

$$= \frac{4.8}{8} = 0.6 \text{ ms}$$

36) Doppler shift

Given

Speed = 150 km/h

frequency = 1.8 GHz

$$150 \times \frac{1000}{3600} = 41.67 \text{ m/s}$$

$$f_d = \frac{41.67 \times 1.8 \times 10^9}{3 \times 10^8} = 250 \text{ Hz}$$

37) share of offered load blocked

Given

System capacity = 250000

Call arrival rate = 70000 calls/min

Avg call duration = 4 min

Offered traffic

$$70000 \times 4 = 280000$$

Blocked calls

$$280000 - 250000 = 30000$$

= 10.71% blocked

38) cluster user capacity

Given

cluster size = 9 cells

users per cell = 600

Total users = $9 \times 600 = 5400$ users

39) calls dropped per second

Given

calls processed = 60000 calls/sec

Drop rate = 0.4%

$$\text{Dropped calls} = \frac{0.4}{100} \times 60000 = 240 \text{ calls/sec}$$

40) Path loss

Given

Transmit Power = 43 dBm

Received Power = -87 dBm

Path loss = $P_t - P_r = 130$ dB

41) Total geographic area, covered

Given

Area covered by one base station = 2.5 km

No. of base stations = 150

Total Area = Area Per station $\times$ No. of station
 $= 2.5 \times 150 = 375$ km²

43) Paging cycles

Given Processed Per Paging cycle = 6000

total subscribers = 150000

Paging cycles = $\frac{\text{Total subscribers}}{\text{users Per cycle}}$

$= \frac{150,000}{6000} = 25$

44) Number of hand offs during a call

Given

call duration = 9 minutes

Hand off every 3 minutes

Hand offs = $\frac{\text{call duration}}{\text{Handoff Interval}} = \frac{9}{3} = 3$

45) signal - to - Noise Ratio (SNR)

Given

signal Power = -85 dBm

Noise Power = -115 dBm

SNR = signal - Noise

$= -85 - (-115) = 30$ dB

46) Minimum band width required

Given

No. of users = 250

Band width per user = 150 kbps

Total band width = 250×150

= 37,500 kbps

= 37.5 Mbps

48) Band width for control channels

Given

No. of control channels = 250

Band width per channel = 12 kHz

Total bandwidth = 250×12

= 3000 kHz

= 3 MHz

49) Total network throughput

Simultaneous connections = 1500

Through put = 1500×600

= 900,000 kbps

= 900 mbps (0.9 Gbps)

50) Number of towers required

Given

coverage radius = 4 km

Area to cover = 350 km²

coverage area of one tower:

$$A = \pi R^2$$

$$= 3.14 \times 4^2 = 50.24 \text{ km}^2$$

$$= 6.97$$

Since a fraction of a tower is not possible